

Predicting Faulty Products in a Casting Process

-Team Members

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GOAL:

The goal of the dataset is to classify whether a casting is defective or not based on various manufacturing parameters such as temperatures, pressure, mold material, operator skill, and more. This helps in identifying the conditions that lead to defects and improving quality control in the casting process.

DATA SET:

- Mold Temperature – Temperature of the mold during casting.
- Pouring Temperature – Temperature of the molten material being poured.
- Cooling Time – Duration for the material to cool.
- Mold Material – Type of mold used (e.g., Graphite, Metal, Sand).
- Cavity Pressure – Pressure inside the mold cavity.
- Mold Age – Age of the mold in use.
- Operator Skill Level – Skill level of the person operating (e.g., Low, Medium, High).
- Alloy Type – Type of alloy used (TypeA, TypeB, etc.).
- Die Wear – Wear level of the casting die.
- Pouring Speed – Speed at which material is poured.
- Defective – Target variable (1 = Defective, 0 = Not Defective).

<https://in.docworkspace.com/d/sII6Dh8AlObPEwAY>

ALGORITHMS USED TO TRAIN inMODELS

- Logistic Regression
- Random Forest Classifier
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Naive Bayes
- Decision Tree Classifier

Reasons Why KNN Performs Best

1. Feature Scaling Helps KNN

- KNN is a distance-based algorithm (usually Euclidean distance).
- Since we scaled the features using StandardScaler, all features contribute equally to the distance calculation.
- This often boosts KNN's performance.

2. Clear Clusters or Similarity Patterns

- If your data has naturally clustered groups (e.g., defect and non-defect parts form tight groups in feature space), KNN can separate them effectively.
- KNN excels when similar items have similar labels, which is common in quality control datasets.

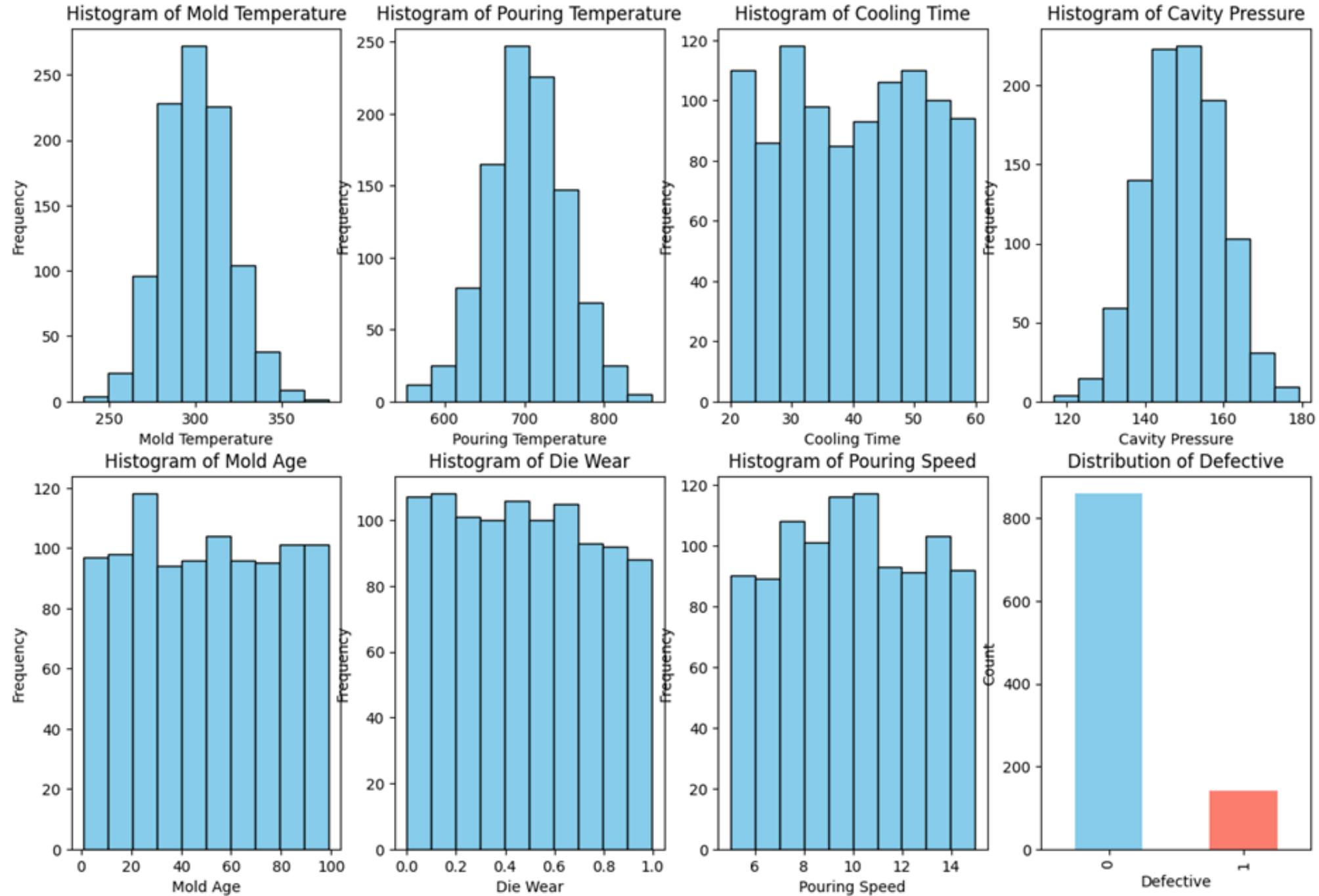
3. Low Dimensionality

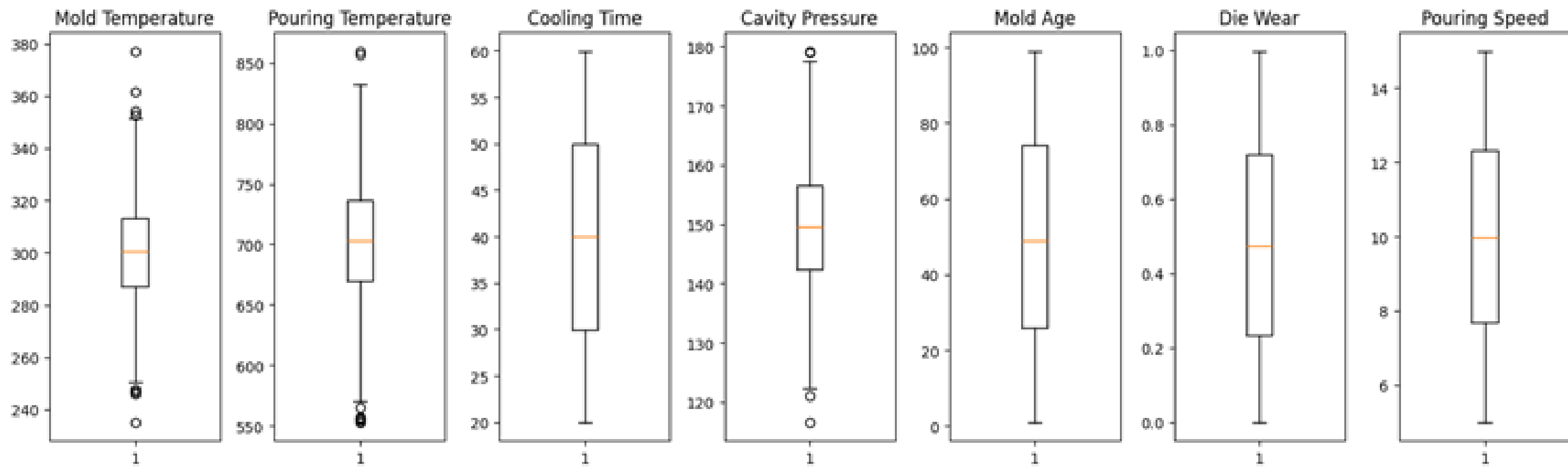
- KNN tends to struggle in high-dimensional spaces (curse of dimensionality).
- If your dataset has relatively few features, this makes KNN more effective.

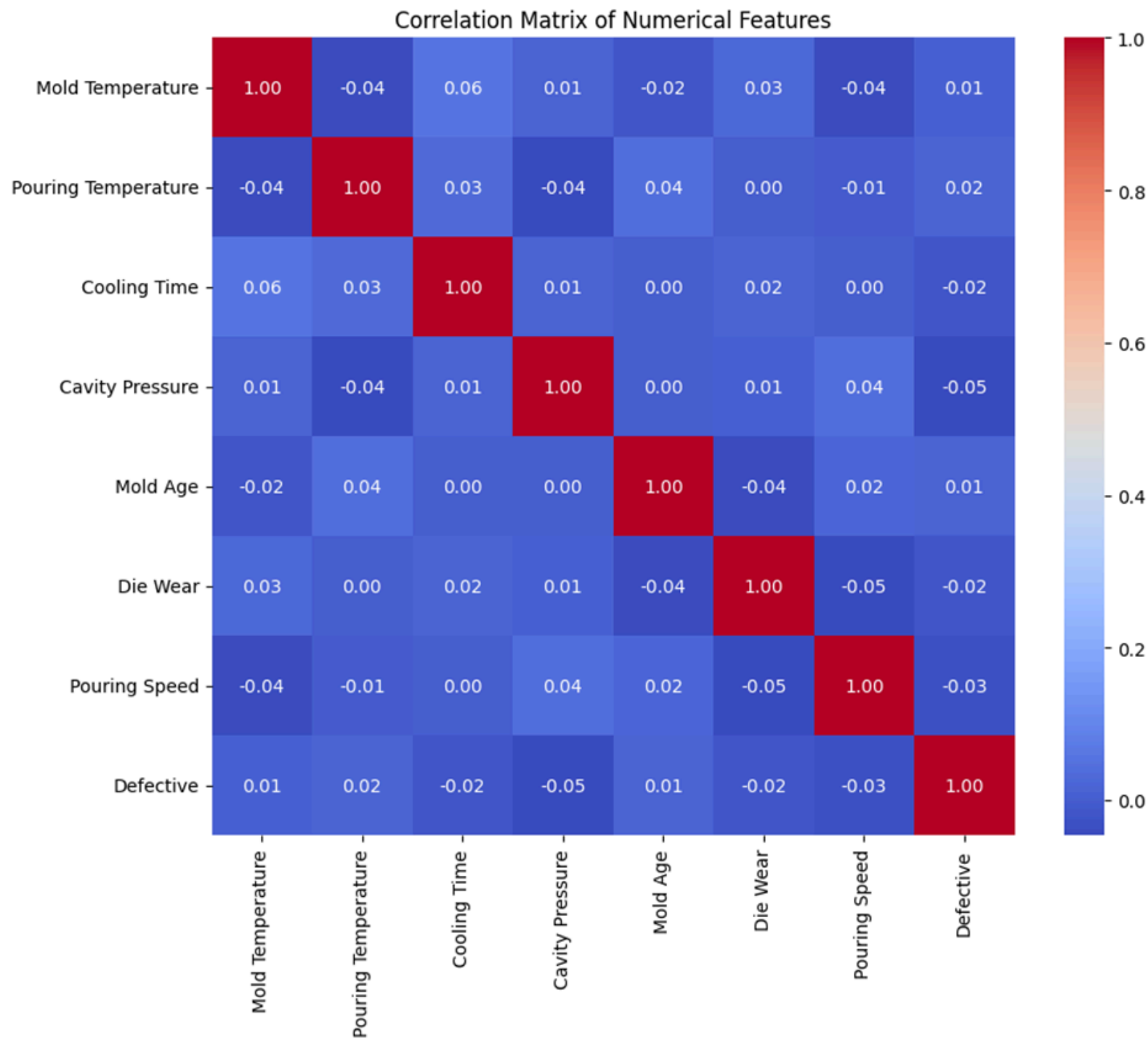
4. Balanced Dataset

- If the target classes (defect vs. non-defect) are balanced, KNN can learn decision boundaries more effectively without bias toward the majority class.
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Data Distribution Analysis







QUESTION AND ANSWERS:

1. What is the best algorithm for classification tasks?

There's no one-size-fits-all answer — the best algorithm depends on:

- Data size
- Number of features
- Linear vs. non-linear relationships
- Imbalance in classes

For example:

- Logistic Regression is great for linearly separable data.
- Random Forest works well on most datasets and handles non-linear data.
- KNN can perform well with well-separated clusters.
- SVM is powerful in high-dimensional spaces.

Best practice: Try several algorithms and compare using metrics like accuracy, F1-score, and cross-validation.

2. How important is feature scaling, and which algorithms need it?

Feature scaling ensures that all features contribute equally to the model, especially for distance-based algorithms.

It's essential for:

- KNN
- SVM
- Logistic Regression
- Gradient Descent-based models

It's not essential for:

- Tree-based models like Decision Trees and Random Forests — they are scale-invariant.

Scaler to use:

- StandardScaler (Z-score normalization) is most common.

Thank
you!